

Haobo Fang

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EDUCATION

University of Michigan - Ann Arbor

Ann Arbor, US

B.Eng. in Computer Engineering, Cumulative GPA: 3.91 / 4.0 Sep. 2022 – Apr. 2026 (expected)

Shanghai Jiao Tong University

Shanghai, China

B.Eng. in Mechanical Engineering, Core GPA: 3.76 / 4.0 Sep. 2022 – Aug. 2026 (expected)

RESEARCH INTERESTS

I am broadly interested in building scalable, adaptable, and reliable robots that interact with humans in daily activities. My research highlights the following perspectives: 1) Origami-Inspired soft robotic arm. 2) Bio-inspired robot. 3) Advanced control system development. 4) Intelligent manufacturing scenarios enhanced by learning.

Research Areas: Soft robotics, Wearable devices, Control system, Reinforcement Learning, Manufacturing

EXPERIENCE

Hybrid Dynamic Robotics Lab

University of Michigan

Undergraduate Researcher, Advisor: Prof. Xiaonan Huang

Sep. 2024 - Present

Origami-Inspired Modular Soft Robotic Arm

- Developed a modular arm based on **Kresling origami actuators**, achieving a predictable linear relationship between contraction and twist.
- Applied the **Piecewise Constant Curvature (PCC) model** for accurate kinematic control with low error ($\approx 2.5\%$).
- Integrated **CAN-based distributed communication and sensing**, enabling plug-and-play modularity, fault tolerance, and self-perception.

Electromagnetic driven Clip Fish

- Leveraged **electromagnetic actuation and bistability** to realize the fin's snap-through motion, enabling a compact and agile robotic fish.
- Employed the **Discrete Elastic Rod (DER) framework** to simulate the tail's motion and optimize its geometric configuration for improved performance.

Data-informed Design and Intelligent Systems Laboratory

Shanghai Jiao Tong University

Undergraduate Researcher, Advisor: Prof. Youyi Bi

Mar. 2023 - May. 2025

A Zero-shot Robot Tool Manipulation Framework based on Visual Prompt-Enhanced Keypoint Representation and Multi-modal Constraints

- Developed a zero-shot robotic manipulation framework leveraging Vision-Language Models (VLMs) and Visual Foundation Models (VFM).
- Integrated task decomposition, keypoint localization, multi-modal primitive inference, and real-time error detection to enable generalizable and precise tool use in complex manufacturing scenarios.

AWARDS

The Jackson and Muriel Lum Scholarship (top 2%)

University of Michigan, 2025

Sep. 2025

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Dean's Honor List	University of Michigan, 2025
Best Poster Award	ICRA 2025 Workshop, 2025
The Jackson and Muriel Lum Scholarship (top 2%)	University of Michigan, 2024
Huatai Securities Technology Scholarship (top 1%)	Shanghai Jiao Tong University, 2024
Top Student in Morality, Intelligence, and Fitness (top 3%)	Shanghai Jiao Tong University, 2023
C-level Scholarship	Shanghai Jiao Tong University, 2023

PUBLICATIONS

- [1] J. Wang, Y. You, X. Zhang, **H. Fang**, J. Wang, and X. Huang, "Origami-Inspired Modular Soft Robotic Arm" *ICRA Workshop on Soft Robotics for Space Applications*, 2025.
- [2] J. Wang, Y. You, X. Xhang, **H. Fang**, J. Wang, and X. Huang, "Origami Inspired Soft Robotic Arm: A Modular Platform for Manipulation" *Extended Abstract, RoboSoft*, 2025.

PATENTS

- [1] J. Wang, Y. You, X. Xhang, **H. Fang**, J. Wang, and X. Huang, "Lightweight, Proprioceptive, Origami-Inspired Soft Robotic Arm for High Payload, Low-Cost Reconfigurable Manipulation," *U.S Patent, Pending*, 2025.

SERVICE

- [1] **Instructional Assistant**, ECE Division, University of Michigan, Fall 2025.
Served as IA for EECS 216: Introduction to Signals and Systems.
- [2] **Volunteer**, IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, May 2025.
- [3] **Presenter**, ICRA workshop, Atlanta, GA, May 2025.
Shared our work in ICRA Workshop on *Soft Robotics for Space Applications*.
- [4] **Captain**, Eighth "Rongchang Chucai" Plan, Shanghai, Spring 2024 – Spring 2025.
Led team activities and coordinated logistics for the eighth cohort.
- [5] **Captain Assistant**, Miyuan Youth Volunteer Group, Shanghai, June 2023 – June 2024.
Supported team operations, participated in 2022 and led 2023 Yunnan volunteer teaching campaign.

COMPETITIONS

China International College Students' Innovation Competition Shanghai Jiao Tong University
Mar. 2024 – Oct. 2024
 Developed an AI-assisted smart teaching platform to enhance education in rural areas.
 Won the 2024 Golden Award in the Shanghai Division.

Liming Cup Mechanical Innovation Competition **Shanghai Jiao Tong University**
Mar. 2023 – Jun. 2023
 Developed a multi-task racing car.
 Won the 2023 Best Design Award and Second Prize.

SKILLS

Python, C/C++, MATLAB (Simulink), Rust, PyTorch, Unity, Solidworks, Wolfram Mathematica, COM-SOL Multiphysics, EasyEDA, STM32, Espressif, Arduino